Predicting E-com Prices

About the data:

- The given dataset has 2452 rows and 8 columns.
- Out of the 8 columns :
    - Item_Rating and Selling_Price are numerical columns.
    - Date is a date field.
    - Product is a unique identifier for each row in the dataset,i.e, there are 2452 products in the data.
    - Product_Brand, Item_Category, Subcategory_1 and Subcategory_2 are categorical columns.

```
Data columns (total 8 columns):
 #   Column         Non-Null Count  Dtype  
---  ------         --------------  -----  
 0   Product        2452 non-null   object 
 1   Product_Brand  2452 non-null   object 
 2   Item_Category  2452 non-null   object 
 3   Subcategory_1  2452 non-null   object 
 4   Subcategory_2  2452 non-null   object 
 5   Item_Rating    2452 non-null   float64
 6   Date           2452 non-null   object 
 7   Selling_Price  2452 non-null   int64  
dtypes: float64(1), int64(1), object(6)
memory usage: 153.4+ KB
```

Data Cleaning and Pre processing :

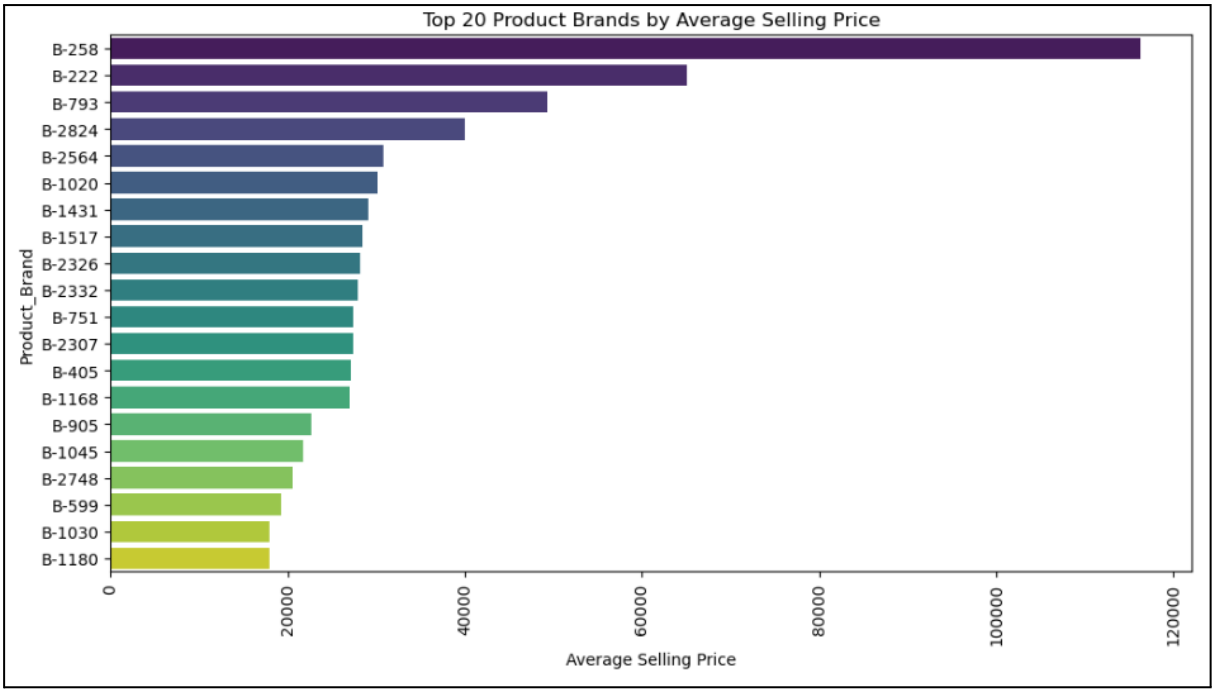
- There are no duplicate rows in the dataset provided.
- There are no null values in the dataset.
- The Date column is changed from Object type to Datetime type.
- New columns for month and year are created.
- The dataset comprises a total of 2,452 individual products, each represented as a unique row, 1,010 distinct product brands, 75 different item categories, 130 unique subcategories within the main item categories, and 336 distinct subcategories within the subcategories of the first level.
- The dataset contains information regarding sales spanning a period of nine years.

Visualisations and Analysis :

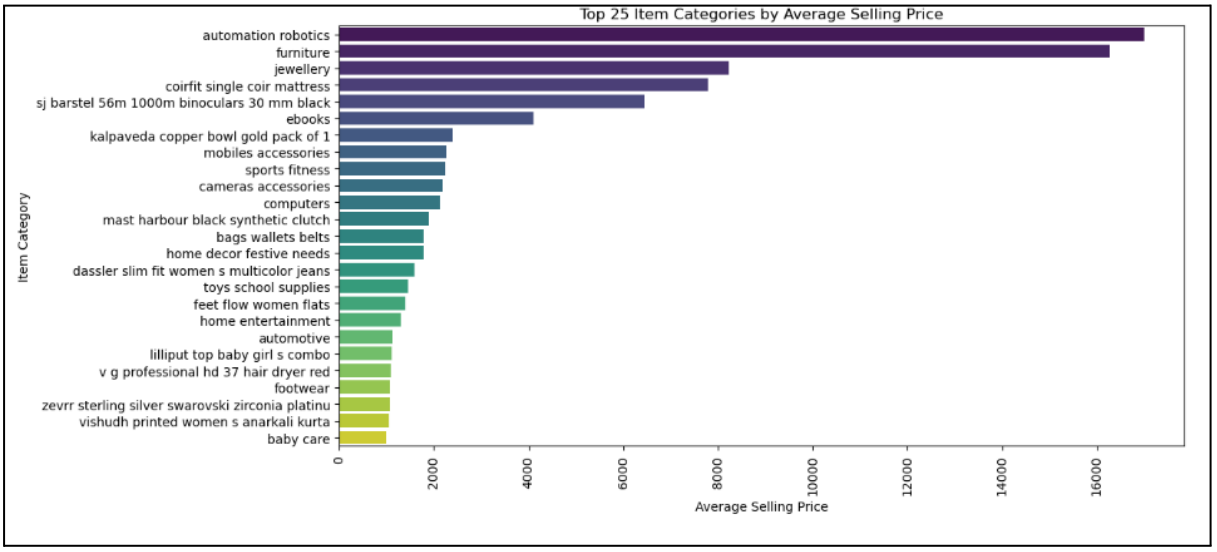
Out of all the plots I plotted in the notebook, the most relevant plots for model selection are:

- Products from Brand B-258 have the highest average selling price, followed by B-222.B-258 might have many products or it might have products with very high prices

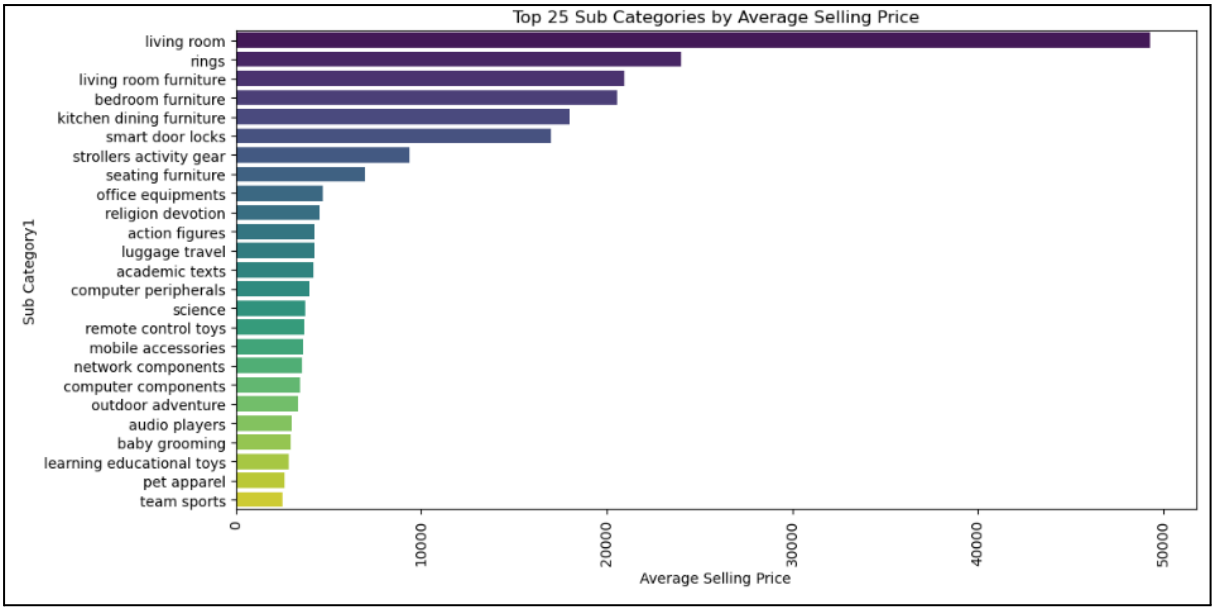
compared to other brands.



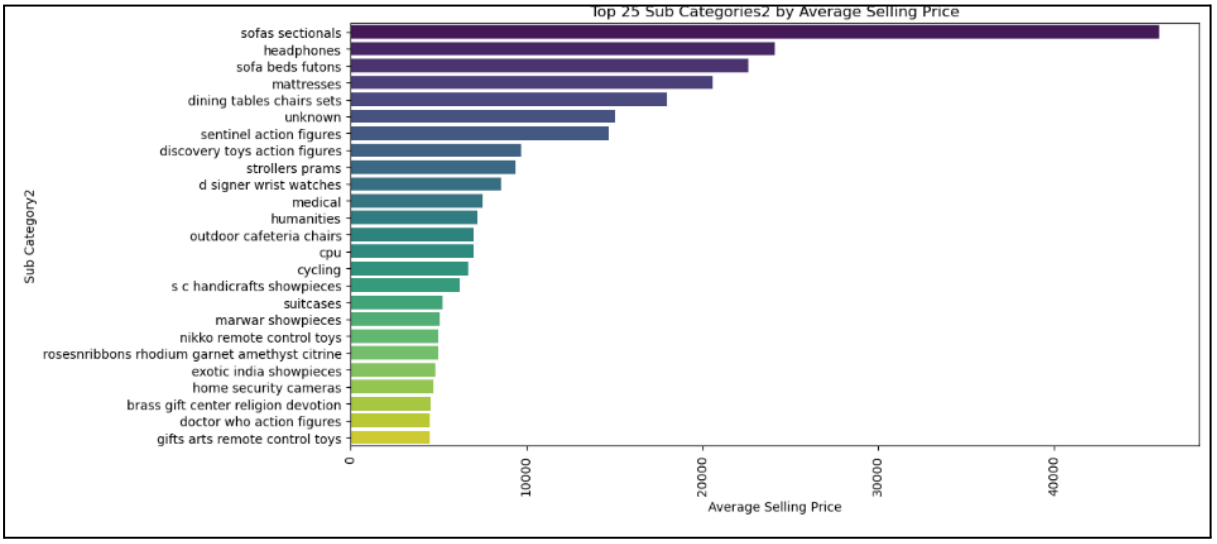
- Automation Robots have the highest average selling price, followed by furniture. It's true that electronic gadgets, furniture and jewellery are the costliest categories of products we use in day to day life.



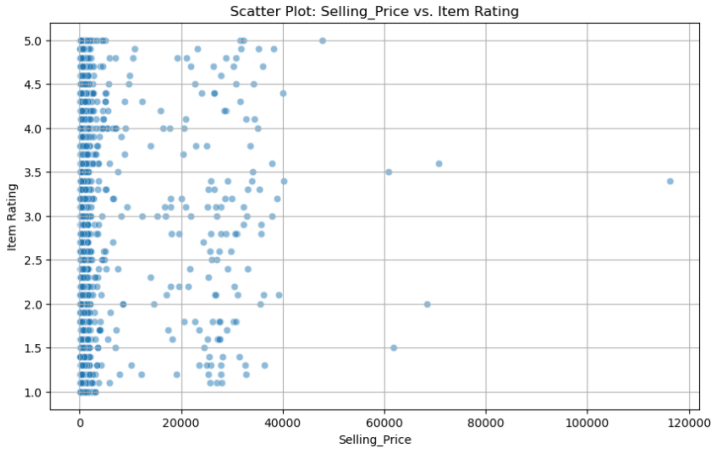
- Products from subcategories Living Room, followed by Rings have the highest average selling prices.



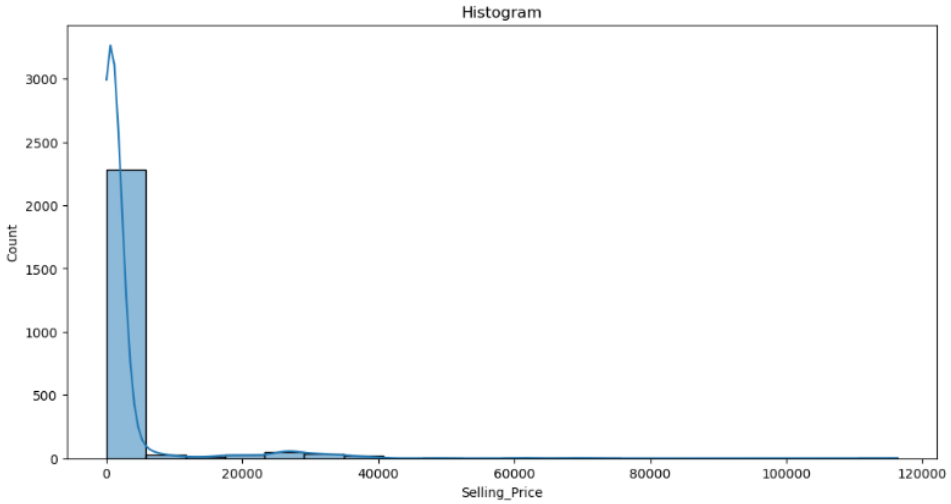
- Products from subcategories_2 Sofa Sectionals, followed by Headphones have the highest average selling prices.



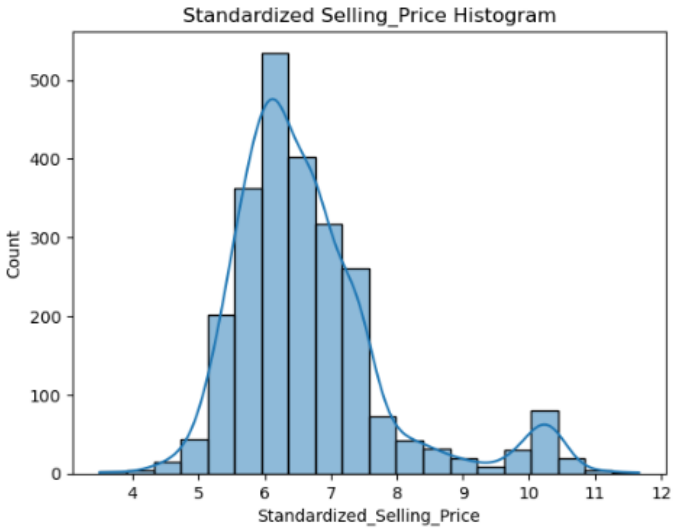
- The scatter plot shows that there isn't a significant effect of 'Item Rating' on 'Selling Price.'



- Selling_price column is highly skewed to the left.This indicates that most of the products fall in the lower price range(Around 10000-15000 USD).Hence,we need to normalise the column.



- Here, log transformation is the best suitable method for normalising the Selling_Price column.



- Since, Product_Brand has 1010 unique values, instead of any kind of encoding,I extracted the numeric part from the values and replaced the column.
- Since the Product field is unique for each row, it won't contribute anything to the model.So, I removed the column.

Encoding Categorical columns:

Approach 1 : One Hot Encoding

- I used One hot encoding for the top 25 values of 'Item_Category', 'Subcategory_1', 'Subcategory_2' based on their average selling price of each entity. This decision was made to manage the size of the dataframe more efficiently. Instead of creating a separate column for each unique value in these categories, I focused on the values which are expected to have more effect on the selling price.

Approach 2 : Target Encoding

- Target encoding all the 3 categorical columns.That is, the average value of the target variable in each level group of the categorical variable is aggregated and replaced with that value.

Model building:

Tried various models like Linear Regression, Random Forest Regressor, Decision Tree Regressor and XGBoost Regressor. Improved performance of Random Forest Regressor and XGBoost Regressor by Best hyperparameters using Grid search CV.

Using One Hot encoding the results of performance are:

Metric	Linear Regression	Random Forest Regressor	Decision Tree Regressor	XGBoost Regressor	Grid Search CV for Random forest	Grid Search CV for XGBoost Regressor
R^2 score	0.434	0.576	0.308	0.519	0.6	0.532
MSE	0.779	0.584	0.952	0.662	0.551	0.645
RMSE	0.883	0.764	0.976	0.814	0.742	0.803

Using Target encoding the results of performance are:

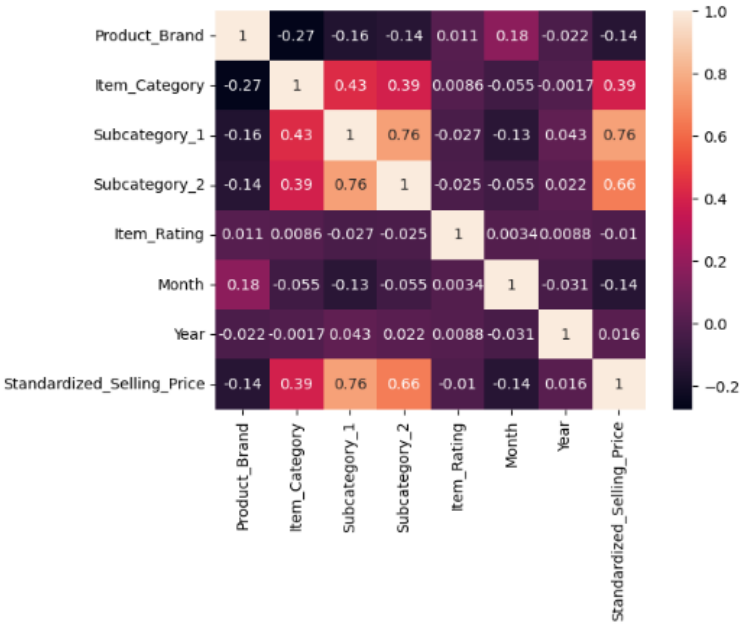
Metric	Linear Regression	Random Forest Regressor	Decision Tree Regressor	XGBoost Regressor	Grid Search CV for Random forest	Grid Search CV for XGBoost Regressor
R^2 score	0.541	0.724	0.501	0.671	0.723	0.712
MSE	0.632	0.38	0.686	0.453	0.382	0.396
RMSE	0.795	0.616	0.829	0.673	0.618	0.63

Importance of Features:

Subcategory_1 is the most influential feature in the model, followed by Subcategory_2 .

```
Subcategory_1: 0.513116341804972
Subcategory_2: 0.20936808867654264
Product_Brand: 0.07510728766720849
Item_Category: 0.07485555163581602
Item_Rating: 0.05350597652122602
Month: 0.041287855496927435
Year: 0.032758898197307346
```

This can also be seen in correlation heatmap.



Conclusion:

Target encoding gave better results than One hot encoding.
After improving performance using hyperparameter tuning, Random Forest Regressor gave the best results:
R^2 score : 0.723
Mean squared error (MSE) : 0.382
Root Mean squared error (RMSE) : 0.618